

FusionDiff: Multi-focus image fusion using denoising diffusion probabilistic models

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ABSTRACT

Multi-focus image fusion (MFIF) is an image enhancement technology with broad application prospects that can effectively extend the depth-of-field of optical lenses. This paper presents a novel MFIF method, named FusionDiff, based on denoising diffusion probabilistic models (DDPM). FusionDiff uses DDPM to fuse two source images by iteratively performing multiple denoising operations. To predict noise accurately and train the model efficiently, a lightweight U-Net architecture is designed as the conditional noise predictor. FusionDiff does not depend on any specific activity level measurement method, fusion rule or complex feature extraction network. It overcomes many algorithm design and training difficulties in existing image fusion methods. FusionDiff is noise-resistant and can still produce outstanding fusion results from source images with noise interference. In addition, FusionDiff is a few-shot learning method, which makes it suitable for image fusion tasks where training samples are relatively scarce. Experiments show that FusionDiff outperforms representative state-of-the-art methods in both visual perception and quantitative metrics. The code is available at <https://github.com/lmn-ning/ImageFusion> <https://github.com/lmn-ning/ImageFusion>

1. Introduction

Depth of field (DOF) is the range that an optical system can clearly image, reflecting the imaging capability of an imaging system. The larger the DOF is, the wider the clear imaging range of the optical system (Li et al., 2020). In practical applications, the finite DOF of an imaging system makes it challenging to capture all-in-focus images; thus, there are often some blurred image areas (Zhang, 2021). As shown in Fig. 1(a)–(d), obtaining globally clear images is a prerequisite for picture beautification, building electronic specimen libraries of insects (Chowdhury, Nguyen, Li, & Hartley, 2021), diagnosing pathology sections (Pei et al., 2021) and performing biomedical experiments. Multi-focus image fusion (MFIF) effectively solves this problem by fusing several locally focused source images. Existing MFIF algorithms can be generally divided into transform domain, spatial domain and deep learning-based methods (Liu, Wang, Cheng, Li, & Chen, 2020).

Transform domain methods include image decomposition-based and feature space transformation-based methods. The image decomposition-based MFIF methods usually include three stages. First, transform the source images into a single-scale (Li, Kang, & Hu, 2013) or multi-scale (Burt & Adelson, 1987; Hui, Manjunath, & Mitra, 2002) space.

Then, perform image decomposition on each scale space. Finally, obtain the fusion result by fusing images with different scales. Moreover, the feature space transformation-based approaches first transform source images into a new feature space and then perform image fusion in the transformed feature space (Mitianoudis & Stathaki, 2007; Tang, Xiong, Yin, Zhu, & Li, 2022). The transform domain methods depend on image decomposition or chunking, thus some vital information is often lost in these processes.

The spatial domain approach regards the MFIF task as a process of selecting clear pixels for image stitching (Li, Kwok, & Wang, 2001). These methods first decompose the source images into several areas based on specific criteria, and then the activity level saliency is calculated for each area (Yu, Liu, & Wang, 2015). Finally, the most significant areas are spliced to produce the fused image. Despite the significant results achieved by these methods, they are susceptible to the region division method, and it is difficult to avoid the formation of artifacts at adjacent region junctions (Liu, Wang, Li, & Chen, 2022).

Despite many achievements of traditional MFIF methods, their development bottlenecks are becoming increasingly apparent. The fusion

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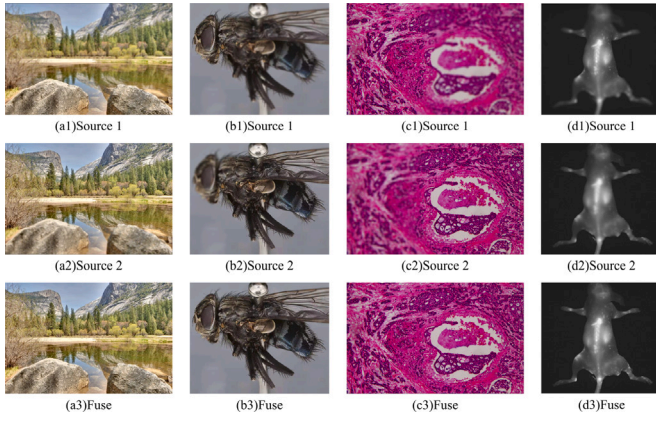


Fig. 1. Examples of application scenarios for MFIF, including natural landscapes (a), macro photography (b), microscope imaging (c) and biomedical experiments (d). The Source-1 and Source-2 denote locally focused source images, and Fuse denotes the desired fused image.

results of these methods are highly influenced by artificial activity level measurement methods and fusion rules (Liu, Liu, & Wang, 2015). To achieve better performance, problems such as measurement methods and design rules have become increasingly complex and difficult. Therefore, traditional MFIF methods encounter increasing challenges in obtaining state-of-the-art (SOTA) results (Liu et al., 2018). Due to the significant breakthroughs of deep learning technology, related deep learning methods have developed as a research hotspot in various image fusion fields (Guo, Fan, Jiang, & Zhang, 2023).

The deep learning-based MFIF methods can be roughly classified into two categories: decision map-based methods and end-to-end methods that do not use decision maps (Zhang, 2021). In the decision map-based MFIF approach, deep learning is typically used to generate a decision map indicating focused regions (Du & Gao, 2017; Liu, Chen, Peng, & Wang, 2017). Then the fusion process is implemented according to the decision map, essentially a spatial domain approach. Thus decision map-based deep learning methods still face difficulties that are difficult to solve in spatial domain methods. End-to-end MFIF methods without decision maps tend to implement MFIF based on extracting and fusing the image features. These methods obtain the fused image through feature extraction, feature fusion and image reconstruction, which relies on the powerful feature extraction capability of neural networks such as convolutional neural network (CNN) and Transformer (Ma et al., 2022; Zhang et al., 2020). In addition, Generative adversarial networks-based (GAN) methods utilize designed adversarial gaming strategies to generate decision maps (Guo et al., 2019) or to force more effective features to be included in the fused image (Zhang, Le, Shao, Xu, & Ma, 2021).

Although MFIF methods have achieved remarkable strides, they all focused on extracting image features or distinguishing clear pixels as fully as possible. In recent years, deep learning MFIF methods that achieve SOTA results use more gain modules to improve the performance (Liu et al., 2022; Pei et al., 2023), which leads to their network structures becoming increasingly large and complex. Scholars must spend much time designing subtle network structures and completing sufficient comparison experiments, which is not conducive to algorithm performance. We reconsider the image fusion task and doubt that the fusion process can be viewed as a generative problem of generating a new image using two source images. Although partial GAN-based methods also consider image fusion as an image generation task, they use the adversarial game mechanism to force the fused image close to the source images rather than modeling it as a purely generative task (Guo et al., 2019; Huang, Le, Ma, Mei, & Fan, 2020; Wang, Xu, Liu, Zhao, & Zhang, 2021). Perhaps the problems existing MFIF methods face could be better addressed from an image generation perspective.

To verify this idea, we propose FusionDiff, a novel MFIF algorithm based on denoising diffusion probabilistic models (DDPM). DDPM (Ho, Jain, & Abbeel, 2020) is a new image generation approach that can easily generate high-quality samples, which has been proven in various computer vision tasks (Lugmayr et al., 2022; Saharia et al., 2023). FusionDiff first uses a series of Gaussian noise to convert a clear image into a pure noise image, and then uses two source images as guidance information to denoise the initial random noise image for several iterations to obtain a fused image. To our understanding, FusionDiff may be the first image fusion algorithm using DDPM. The main contributions of this work are summarized as follows:

- We present a DDPM-based MFIF method named FusionDiff, and achieve SOTA fusion results.
- FusionDiff differs from existing MFIF methods from the fusion principle perspective, independent of any manual activity level measures, fusion rules, or complex feature extraction networks. This means designing and training FusionDiff is simpler and easier to improve performance.
- FusionDiff is a few-shot learning approach that uses less than 2% of the training data of existing MFIF algorithms to make FusionDiff work well for MFIF tasks.
- FusionDiff has some noise robustness, i.e., satisfactory fusion results can be obtained even when noise interference exists in the source images.

The rest of this paper is unfolded as follows. Section 2 introduces previous works related to MFIF and DDPM. Section 3 details the proposed method, and Section 4 shows the experimental results and analysis. Section 5 summarizes the paper.

2. Related work

2.1. Multi-focus image fusion methods

2.1.1. Transform domain methods

Transform domain methods are the earlier appearing image fusion methods, including image decomposition-based methods and feature space transform-based methods. The former is to decompose the source image in the base and detail layers at multiple scales (Jiang et al., 2020; Li et al., 2013), which is in line with human visual cognitive characteristics and can provide relevant information such as human visual sensitivity contrast. Classical image decomposition methods include pyramid transform (Burt & Adelson, 1987), discrete wavelet transform (Hui et al., 2002) and multi-scale geometric analysis (da, Arthur, Jianping, & Minh, 2006).

Feature space transform-based methods usually use advanced image representation theory to obtain new feature spaces and use sliding windows to obtain translation invariance. Feature space transformation methods that have proven effective include sparse representation (Tang et al., 2022; Yang & Li, 2010), independent component analysis (Mitianoudis & Stathaki, 2007), high-order singular value decomposition (Liang, Wang, Ding, & Zeng, 2012), convolutional sparse representation (Yu, Xun, Ward, & Wang, 2016), etc. In addition, NSCT-SR (Liu et al., 2015) combines the advantages of multiscale decomposition and feature space transformation to achieve good fusion results.

2.1.2. Spatial domain methods

Spatial domain methods perform image fusion directly in the pixel space of the image based on the features of pixel points or blocks of pixels. These methods have a simple principle and are easy to execute. The simplest method is to directly compute the mean image of all the source images to obtain the fused image, but such a simple superposition leads to blurred edges and contours in the fused image (Liu et al., 2020). In 2001, Li et al. (2001) proposed an image chunking-based

MFIF method. They used spatial frequency to measure the activity level of each image block, but the block size significantly impacts its fusion effect. Afterward, they improved the MFIF method by using an artificial neural network to classify the features of each image block and thus determine its focusing properties (Li, Kwok, & Wang, 2002). Since then, the image chunking-based approach has become popular in MFIF. To obtain high-quality decision maps, scholars have conducted numerous studies on two key issues: activity level measurement (Chen, Li, Luo, & Ma, 2022; Yu et al., 2015; Zhan, Li, Teng, Wang, & Shi, 2015) and regional division strategy (Guo, Zhang, & Zhou, 2014; Hariharan, Koschan, & Abidi, 2007). Although many methods for measuring the focusing properties of image blocks have emerged, how to deal with the intersection of focused and defocused regions is a problem faced by almost all spatial domain methods.

2.1.3. Deep learning-based methods

Deep learning is widely used to generate weight maps of fused images or to directly extract features from source images for fusion, corresponding to decision map-based and end-to-end methods (Zhang, 2021), respectively.

In 2017, Liu et al. (2017) used a CNN to determine whether each image patch belonged to focused regions through binary classification, then assigned different weights to these patches to guide the fusion process. Following (Liu et al., 2017), various efforts are proposed to improve the classification accuracy of CNN, including multi-scale input (Du & Gao, 2017), pixel-level focus attribute judgment (Tang, Xiao, Li, & Wang, 2018; Wang, Zhao, Ren, Xu, & Yu, 2020), ensemble learning of multiple CNNs (Amin-Naji, Aghagolzadeh, & Ezoji, 2019), self-supervised learning (Wang, Li, Duan, & Zhang, 2022), discarding patch operation (Li et al., 2020), designing complex network architectures (Liu et al., 2022), and the use of attention mechanisms (Guo, Meng, Mei, Weng, & Tong, 2020) and complex loss functions (Cheng, Xu, & Wu, 2023; Ma et al., 2021). However, most of these algorithms necessitate extensive post-processing for the decision map, such as guided filtering, morphological operations, the watershed algorithm, small area removal, etc., before they can be used for image fusion. In addition to CNNs, GANs are widely used for MFIF tasks. FuseGAN (Guo et al., 2019) forces the generator to construct a decision map accurately by building an adversarial game relationship between positive and negative samples. Based on FuseGAN, MFIF-GAN (Wang et al., 2021) introduces prior knowledge of the focus region to perfect the decision map. Furthermore, Hu, Jiang, Liu, and Ma (2023) implemented a zero-shot learning MFIF algorithm (ZMFF) based on the deep image prior (DIP) method (Ulyanov, Vedaldi, & Lempitsky, 2018) to address the lack of high-quality MFIF datasets.

End-to-end MFIF methods tend to implement MFIF based on extracting and fusing the image features. Xu, Qin, Wang, Zhang, and Ye (2018) proposed the first end-to-end MFIF algorithm using a two-stream network in which one branch processes one source image. The method obtains the fused image through feature extraction, fusion and reconstruction. The IFCNN (Zhang et al., 2020) has a comparable architecture, but its feature fusion process differs from previous methods. In addition, the fusion principle of the MFIF algorithm based on a one-stream network proposed by Li et al. (2019) is similar to that in Xu et al. (2018). GANs are also used for end-to-end MFIF tasks. ACGAN (Huang et al., 2020) takes the different gradient distributions of two input images as prior information to improve image quality. MFF-GAN (Zhang et al., 2021) and GIDGAN (Zhou, Hou, Zhang, Ma, & Ling, 2022) use the gradient loss function to reduce the chromatic aberration between the source and fused images. Although an adversarial mechanism using GANs helps to extract more information from source images, this information is challenging to describe in terms of traditional mathematical forms of loss functions. This results in GAN-based MFIF algorithms struggling to improve the fusion quality from the perspective of multiple image features. In addition, SwinFusion (Ma et al., 2022) and TransFusion-Net (Pei et al., 2023) used Transformer to

design fusion networks for cross-regional complementary information interaction and extraction, resulting in high-quality image fusion from a global perspective. To fully extract the information from source images, end-to-end MFIF methods usually employ complex loss functions, such as perceptual (Li et al., 2019; Zhang et al., 2020), structural similarity (Huang et al., 2020; Ma et al., 2022; Xu, Ma, Jiang, Guo, & Ling, 2022), gradient (Huang et al., 2020; Zhang et al., 2021), intensity (Ma et al., 2022; Zhou et al., 2022) and texture (Ma et al., 2022) loss functions.

2.2. Denoising diffusion probabilistic models

The denoising diffusion probabilistic models (DDPM) (Ho et al., 2020) consists of two Markov chains, a forward process and a reverse process. The forward process converts the initial sample F_0 to pure noise $F_T \sim N(0, I)$ by adding Gaussian noise to F_0 in T steps, and the reverse process gradually denoises the input random noise F_T to produce a clear output image F_0 . Each forward step is defined as:

$$q(F_t|F_{t-1}) = N(F_t; \sqrt{1-\psi_t}F_{t-1}, \psi_t I) \quad (1)$$

where the variance ψ_t increases gradually with time step t . To speed up the computation, F_t can be directly sampled from F_0 without t iterations in the forward process. Define $\phi_t = 1 - \psi_t$ and $\bar{\phi}_t = \sum_{s=1}^t \phi_s$. The forward process is expressed as:

$$q(F_t|F_0) = N(F_t; \sqrt{\bar{\phi}_t}F_0, (1 - \bar{\phi}_t)I) \quad (2)$$

Eq. (2) implies that the distribution of F_t is closely related to F_0 , and the effect of F_0 on F_t decreases as t increases.

Since the random sampling operation is non-differentiable, the gradient descent method cannot be used for training. A common alternative method is to use the re-parameterization trick (Kingma & Welling, 2013) to make the sampling process differentiable. Specifically, sampling $z \sim N(\mu, \sigma^2)$ can be written as $z = \mu + \sigma\epsilon$, where ϵ is a random variable obeying the Gaussian distribution. Eq. (2) is written using the re-parameterization trick as:

$$F_t = \sqrt{\bar{\phi}_t}F_0 + \sqrt{1 - \bar{\phi}_t}\epsilon_t, \text{ where } \epsilon_t \sim N(0, 1 - \phi_1\phi_{t-1} \dots \phi_1) \quad (3)$$

The inverse process of the DDPM refers to obtaining F_{t-1} through F_t , which is the opposite of Eq. (1), and each inverse sampling step also obeys the Gaussian distribution:

$$p_\theta(F_{t-1}|F_t) = N(F_{t-1}; \mu_\theta, \Sigma_\theta) \quad (4)$$

where the variance Σ_θ is a constant related to ψ_t and the mean μ_θ is obtained by neural network learning. The DDPM training objective is to learn Eq. (4), and its objective function is:

$$L_t = E_{F_0, t, \epsilon_t} (\|\epsilon_t - \epsilon_\theta(F_t, t)\|^2) \quad (5)$$

where ϵ_θ is the noise predictor with F_t and t as inputs, whose goal is to predict the noise added in the forward process. According to the predicted noise $\epsilon_\theta(F_t, t)$, Eq. (4) is written using the re-parameterization trick as:

$$F_{t-1} = \frac{1}{\sqrt{\phi_t}} \left(F_t - \frac{1 - \phi_t}{\sqrt{1 - \bar{\phi}_t}} \epsilon_\theta(F_t, t) \right) + \sqrt{\Sigma_\theta} \epsilon, \text{ where } \epsilon \sim N(0, I) \quad (6)$$

The DDPM implements the inverse process according to Eq. (6), i.e., the image generation process. All variables in Eq. (6) except $\epsilon_\theta(F_t, t)$ are known, so the key to the inverse process is to predict $\epsilon_\theta(F_t, t)$ accurately, which means that a strong noise predictor is necessary for the DDPM.

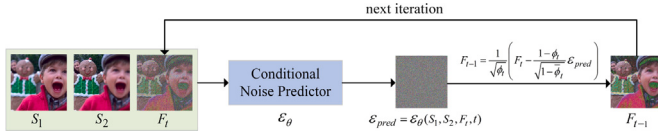


Fig. 2. Demonstration of the image fusion principle by FusionDiff, where S_1 and S_2 denote the right-focused and left-focused source image, respectively.

3. Proposed method

3.1. FusionDiff

Unlike image generation tasks in general, image fusion is a relatively deterministic process. In MFIF, it is easy for observers to distinguish which pixels of each source image are focused to roughly determine which regions should be included in the fused image. In other words, for a series of source images, the generated results of FusionDiff should be consistent with the expected fusion results of the observer rather than randomly generated. Thus FusionDiff does not add any random noise in the image generation process to keep away from the randomness of the fused image. Different from Eq. (6), the reverse process of FusionDiff is defined as:

$$F_{t-1} = \frac{1}{\sqrt{\phi_t}} \left(F_t - \frac{1 - \phi_t}{\sqrt{1 - \phi_t}} \varepsilon_\theta(S_1, S_2, F_t, t) \right) \quad (7)$$

The experimental results in Table 6 show that the MFIF process based on Eq. (6) is unstable, while Eq. (7) can solve this problem well. More importantly, the MFIF algorithm based on Eq. (7) can obtain better fusion results. Moreover, this paper follows the objective function of the DDPM, i.e.

$$L_t = E(\|\varepsilon_t - \varepsilon_\theta(S_1, S_2, F_t, t)\|^2) \quad (8)$$

Fig. 2 illustrates the MFIF principle of FusionDiff. In the reverse process, two source images S_1 , S_2 , and latent variable F_t are input into the conditional noise predictor ε_θ together. The goal of ε_θ is to predict the noise $\varepsilon_\theta(S_1, S_2, F_t, t)$ added in each forward diffusion step, and then generate F_{t-1} that is clearer than F_t by Eq. (7). After many iterations, F_0 is obtained as the final fusion result. In this subsection, the architecture of ε_θ , the variance of the forward process, and the training and inference processes are described in detail.

3.1.1. Conditional noise predictor

As shown in Fig. 3, U-Net is used as the backbone of the noise predictor. FusionDiff can theoretically fuse any number of images simultaneously, and this paper uses the fusion of two images as an example. To make S_1 and S_2 to be the guiding conditions, S_1 , S_2 and F_t are spliced in the channel dimension as the input of ε_θ , and then the channel count of the feature map is converted to c through a convolution module. ε_θ includes a down-sampling module, a middle module and an up-sampling module. The down-sampling and up-sampling modules contain four steps, consisting of several residual blocks, two down-sampling layers and two up-sampling layers. This paper uses a two-dimensional convolution with a step size of 2 to implement down-sampling, and a two-dimensional transposed convolution (Noh, Hong, & Han, 2015) is used to implement up-sampling. Additionally, the output of each down-sampling step is added to the input of the corresponding up-sampling step (He, Zhang, Ren, & Sun, 2016) to take full advantage of image features. For the time step t , the position embedding in the Transformer (Vaswani et al., 2017) is used to convert t to the time embedding t_e , and t_e is added to each residual block. Notably, the noise predictor in this paper is lightweight compared to that in the DDPM (Ho et al., 2020), which can significantly reduce the training cost. As shown in Table 5, compared to DDPM,

FusionDiff reduces the network size from 35.71M to 26.90M and FLOPs (floating-point operations per second) from 1514.80G to 239.86G.

In this paper, the influence of the splicing order of S_1 , S_2 and F_t on the fusion result when they are input to ε_θ is considered. There are three splicing modes: $S_1 S_2 F_t$, $S_1 F_t S_2$ and $F_t S_1 S_2$. The experiments show that different input types have little effect on the fusion results, so the easiest form, $S_1 S_2 F_t$, is adopted.

3.1.2. Forward process variance

The variance ψ_t in the forward process is one of the most important parameters for FusionDiff. According to the analysis by Nichol and Dhariwal (Nichol & Dhariwal, 2021), ψ_t adopts a cosine form and is defined as:

$$\psi_t = 1 - \frac{f(t)}{f(t-1)}, \text{ where } f(t) = \cos\left(\frac{t/T + s}{1 + s} \cdot \frac{\pi}{2}\right)^2 \quad (9)$$

where $s = 0.008$, and the total diffusion step $T = 2000$ is set in this paper. In FusionDiff, the forward process of gradually changing from a clear image to a pure noise image is shown in Fig. 4.

3.1.3. Training

Algorithm 1 displays the training process of FusionDiff. Two input source images S_1 , S_2 , an all-in-focus fused image F and the total diffusion step T are used to train FusionDiff. The conditional noise predictor ε_θ is randomly initialized before training. First, the image pair (S_1, S_2, F) is randomly selected from the training dataset, the time step t is randomly sampled from the integer set $\{1, \dots, T\}$, and a random noise ε_t is sampled from the Gaussian distribution. Then, the conditional noise predictor is optimized by Eq. (7) and the gradient descent strategy.

Algorithm 1 Training.

Input: Two source images S_1 , S_2 , an all-in-focus fused image F , and total diffusion step T
Initialize: Randomly initialized the conditional noise predictor ε_θ
repeat
 Sample (S_1, S_2, F) from the training dataset
 Sample $t \sim \text{Uniform}(\{1, \dots, T\})$
 Sample $\varepsilon_t \sim N(0, 1 - \phi_t \phi_{t-1} \dots \phi_1)$
 Calculate the gradient using $\nabla_\theta \|\varepsilon_t - \varepsilon_\theta(S_1, S_2, F_t, t)\|^2$
until converged

3.1.4. Inference

In the inference phase, as illustrated in Algorithm 2, S_1 and S_2 are input to FusionDiff. In each inference step, S_1 and S_2 are input into the conditional noise predictor. The initial hidden variable F_T is sampled from the Gaussian distribution, and the inverse iteration starts from $t = T$. Each iteration generates a latent variable F_{t-1} of the fused image with different noise levels. As t decreases, the lower the noise level is, the clearer the fused image. And F_0 is returned as the final fusion result at the end of the iteration.

Algorithm 2 Inference.

Input: Two source images S_1 , S_2 , and total diffusion step T
Load: Conditional noise predictor ε_θ
 Sample $F_T \sim N(0, I)$
for $t = T, T-1, \dots, 1$ **do**
 Calculate $F_{t-1} = \frac{1}{\sqrt{\phi_t}} \left(F_t - \frac{1 - \phi_t}{\sqrt{1 - \phi_t}} \varepsilon_\theta(S_1, S_2, F_t, t) \right)$
end for
return F_0 as the fused image

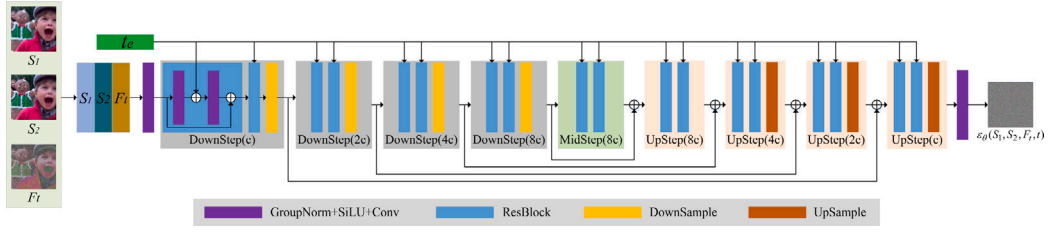


Fig. 3. The architecture of conditional noise predictor architecture, where S_1 and S_2 denote the right-focused and left-focused source image, respectively.



Fig. 4. Demonstration of the forward process of FusionDiff.

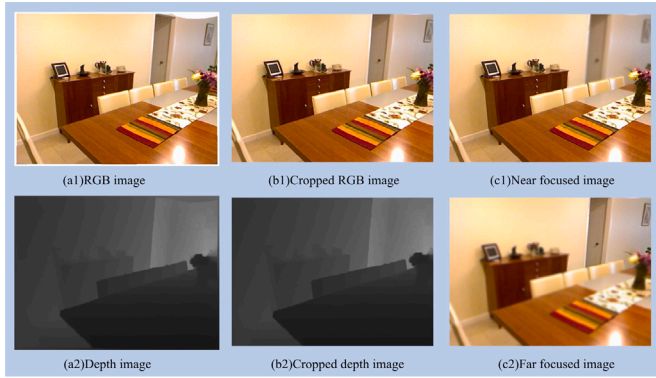


Fig. 5. Demonstration of generating multi-focus image pairs (c) from RGB-D images (a).

4. Experiments

In this section, the performance of the proposed method is verified through extensive experiments. First, the experimental settings are introduced, including the datasets, training details and eight objective evaluation indexes used. Then FusionDiff is compared with thirteen representative SOTA methods using publicly available datasets, and qualitative and quantitative evaluation results are provided for all methods, demonstrating that FusionDiff is an excellent few-shot learning image fusion method. Finally, experiments show that the fusion process of FusionDiff is highly deterministic and noise-robust.

4.1. Experimental settings

4.1.1. Datasets

Due to the optical properties of an imaging lens, the clear and blurry areas of an image are typically correlated with the depth of the scenes. Therefore, a reasonable way to create an MFIF dataset is to generate focused and defocused image pairs from an RGB-D dataset containing image depth information. Based on Zhang et al. (2020), 100 RGB images are chosen randomly from the NYU-D2 dataset, and the corresponding depth map is used to generate an MFIF dataset. In particular, each image of the NYU-D2 dataset has a size of 640×480 pixels and is surrounded by a white box approximately 20 pixels wide, and the same is true for the obtained MFIF dataset. The experiments show that the direct use of these images for training leads to poor fusion results of the marginal part in the fused image. This is because the MFIF model cannot learn information conducive to image fusion from

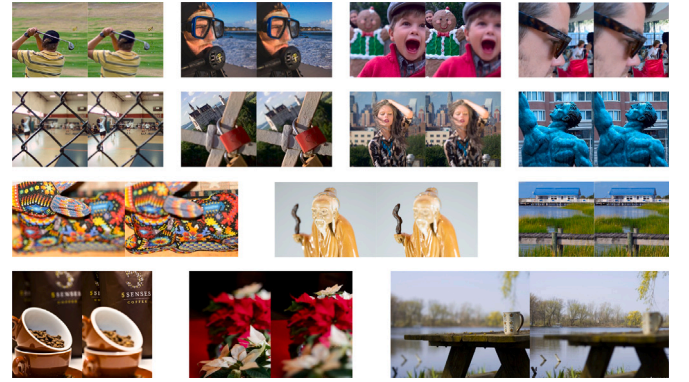


Fig. 6. Partial multi-focus image pairs from the Lytro and MFFW dataset.

Table 1

The number of training image pairs for various deep learning-based MFIF algorithms.

Method	Backbone	Training dataset size
CNN	CNN	1,000,000
p-CNN	CNN	1,450,000
ECNN	Ensemble of CNNs	1,000,000
DRPL	CNN	20,000
SESF-Fuse	CNN	82,783
MUFusion	CNN	21,780
SDNet	CNN	184,790
FuseGAN	GAN	5,850
MFIF-GAN	GAN	11,355
IFCNet	CNN	100,000
ACGAN	GAN	23,714
MFF-GAN	GAN	22,090
DRL-FPD	CNN	31,775
TransFusion-Net	Transformer	18,000
SwinFusion	Transformer	29,595
FusionDiff	DDPM	100

pure white regions without any content. Hence these white box regions are cropped, and the final image size is 600×440 pixels. The process of obtaining multi-focus image pairs from RGB-D images is shown in Fig. 5. Tests are performed using the publicly available Nejati, Samavi, and Shirani (2015) and MFFW (Xu, Wei, Zhang, Liu, & Zhang, 2020) datasets to validate the performance of FusionDiff, where the Lytro dataset contains 20 pairs of images with a resolution of 520×520 , and the MFFW dataset contains 13 pairs of images of different sizes. Partial images from the test dataset are displayed in Fig. 6.

Noteworthy, FusionDiff is a few-shot learning model, meaning that much effort is not required to produce large-scale datasets. The fusion

results obtained by FusionDiff are independent of a large amount of training data, and FusionDiff realizes the transformation from data-driven to model-driven. Table 1 lists the training dataset size of several representative deep learning image fusion algorithms, including CNN (Liu et al., 2017), p-CNN (Tang et al., 2018), ECNN (Amin-Naji et al., 2019), DRPL (Li et al., 2020), SESF-Fuse (Ma et al., 2021), MUFusion (Cheng et al., 2023), SDNet (Zhang & Ma, 2021), FuseGAN (Guo et al., 2019), MFIF-GAN (Wang et al., 2021), IFCNN (Zhang et al., 2020), ACGAN (Huang et al., 2020), MFF-GAN (Zhang et al., 2021), DRL-FPD (Liu et al., 2022), TransFusion-Net (Pei et al., 2023) and SwinFusion (Ma et al., 2022). FusionDiff can achieve good fusion results as the other algorithms use less than 2% of the training data they use, which will significantly improve the dependence of the fusion model on the dataset.

4.1.2. Training details

It takes 100,000 epochs to train, and the batch size is 32. The AdamW optimizer is used in this paper, and the learning rate initialized to 0.0002 and declined by a factor of 0.99 every 1000 epochs. For the conditional noise predictor, the number of channels is set to $c = 64$, and the num-groups of GroupNorm are 32. Eq. (8) is used to train the conditional noise predictor, and to prevent the network weight from stopping updating due to the excessively small gradient value, the loss is amplified 128 times during the backward phase because the loss value in the training process is relatively small. In the conditional noise predictor, the sizes of the up-sampling and down-sampling feature maps of the residual connection are aligned, so FusionDiff supports images with arbitrary sizes as input. The image size is 256×256 pixels during the training phase, and it does not need to be modified during the inference phase. In addition, we implement the proposed method using the PyTorch framework. And all experiments are implemented on the i7-12700 CPU and NVIDIA RTX 3090 GPU.

4.1.3. Quality evaluation metrics

The fusion results are compared using quantitative and qualitative evaluations to objectively evaluate the performance of different methods. The quantitative comparison refers to using some image quality evaluation metrics to assess the quality of the fusion results. Generic image fusion metrics can be classified into five types: information theory-based, image feature-based, image structural similarity-based, human perception inspired, and image correlation-based metrics. In this paper, eight popular evaluation metrics are selected from these types, i.e., the peak signal-to-noise ratio (PSNR) (Jagalingam & Hegde, 2015), edge-based similarity measurement ($Q_{AB/F}$) (Xydeas et al., 2000), structural similarity index measure (SSIM) (Wang, Bovik, Sheikh, & Simoncelli, 2004), visual information fidelity for fusion (VIFF) (Han, Cai, Cao, & Xu, 2013), correlation coefficient (CC) (Jagalingam & Hegde, 2015), information entropy (EN) (Roberts, Van Aardt, & Ahmed, 2008), nonlinear correlation information entropy (Q_{NCIE}) (Wang, Shen, & Zhang, 2005) and gradient-based metric (Q_G) (Xydeas et al., 2000). Especially, these eight metrics are positive indicators; that is, the higher the score, the better the fusion result.

4.2. Comparison with SOTA methods

4.2.1. Quantitative comparison

In this subsection, FusionDiff is compared with thirteen classical SOTA algorithms, including NSCT-SR (Liu et al., 2015), GFF (Li et al., 2013), CSR (Yu et al., 2016), DSIFT (Yu et al., 2015), CNN (Liu et al., 2017), U2Fusion (Xu et al., 2022), MUFusion (Cheng et al., 2023), MFIF-GAN (Wang et al., 2021), ZMFF (Hu et al., 2023), IFCNN (Zhang et al., 2020), SDNet (Zhang & Ma, 2021), MFF-GAN (Zhang et al., 2021) and SwinFusion (Ma et al., 2022), using the evaluation metrics described in Section 4.1.3. According to the introduction in Section 2, these methods basically include all types of current MFIF algorithms, and their algorithm categories are shown in Table 2.

Table 2

Categories, implementation methods and backbone networks of thirteen SOTA methods.

Method	Category	Implementation method	Backbone
NSCT-SR	Transform domain	Multi-scale decomposition	
GFF	Transform domain	Single-scale decomposition	
CSR	Transform domain	Feature space transform	
DSIFT	Spatial domain	Decision map-based	
CNN	Deep learning	Decision map-based	CNN
U2Fusion	Deep learning	Decision map-based	CNN
MUFusion	Deep learning	Decision map-based	CNN
MFIF-GAN	Deep learning	Decision map-based	GAN
ZMFF	Deep learning	Decision map-based	CNN with DIP
IFCNN	Deep learning	End-to-end	CNN
SDNet	Deep learning	End-to-end	CNN
MFF-GAN	Deep learning	End-to-end	GAN
SwinFusion	Deep learning	End-to-end	Transformer

Tables 3 and 4 display the evaluation scores of FusionDiff and other SOTA methods using the eight evaluation metrics on the Lytro and MFFW datasets, respectively. The average scores of each method are listed, with the highest values highlighted in bold and the second values underlined. It can be seen that FusionDiff ranks in the top two in all evaluation scores, indicating that the fusion results produced by FusionDiff have a high correlation with the corresponding source images and contain the most valid information, the least distortion in terms of the image structure, and the highest fidelity of human visual information. Overall, FusionDiff outperforms other comparison methods in objective evaluation.

4.2.2. Qualitative comparison

In this subsection, two representative test image pairs from the Lytro and MFFW dataset are used to visually compare the fusion results generated by FusionDiff and other SOTA methods.

Fig. 7(a)–(p) shows the “Souvenir in front of Sydney Opera House” source image pair from the Lytro dataset and the corresponding fused images produced by various MFIF methods. Furthermore, in order to compare the performance of the different algorithms more clearly, Fig. 7 also shows the difference maps of each fused image with the first source image (a). These fusion results are compared by the detailed parts of the koala’s right ear and human finger, which are marked and enlarged with red and green frames, respectively. The focused and defocused region distributions in these two areas are complex, resulting in a difficult fusion process, and thus, the performance of different MFIF methods can be thoroughly compared. In the fused images of NSCT-SR(c), GFF(d), CSR(e), DSIFT(f), CNN(g), U2Fusion(h), MUFusion(i), IFCNN(l), SDNet(m) and SwinFusion(o), the right ear of the koala is significantly blurred, which is caused by the difficulty these methods have in accurately distinguishing the focused and defocused features in complex multi-focus scenes. The differences between these fused images can be observed more clearly by the circular textures in the red boxes of the difference maps. In the green box area, consistent with the distribution of source images, all the fused images should show a distinct outline of the finger and a larger black part of the souvenir. However, the contour of the finger is blurry in the results of NSCT-SR(c), GFF(d), CSR(e), U2Fusion(h), MUFusion(i), ZMFF(k), IFCNN(l), SDNet(m), MFF-GAN(n) and SwinFusion(o). And the black part of the souvenir is missing in the fusion results of DSIFT(f), CNN(g) and MFIF-GAN(j). Only FusionDiff(p) shows good fusion results in all these regions, proving the superiority of FusionDiff in complex multi-focus scenes.

Fig. 8(a)–(p) shows the “old man with a stick” source image pair from the MFFW dataset and the corresponding fused images produced by various MFIF methods. Areas of clothing with complex textures in the blue boxes are enlarged and displayed in the bottom right corner of each image. Although all methods produce acceptable results, differences are observed in the areas indicated by the blue boxes. Comparing (a) and (b) reveals that the blue box belongs to the focused region,

Table 3

The average scores of all algorithms on the Lytro dataset, where the highest and second highest scores are highlighted in bold and underlined, respectively.

Methods	<i>PSNR</i>	<i>Q_{AB/F}</i>	<i>SSIM</i>	<i>VIFF</i>	<i>CC</i>	<i>EN</i>	<i>Q_{NCIE}</i>	<i>Q_G</i>
NSCT-SR	74.6930	0.5534	0.8539	1.3978	0.9717	7.6813	0.8391	0.7498
GFF	74.7037	0.5641	0.8536	1.4224	0.9718	7.6821	0.8375	0.7501
CSR	74.6310	0.5799	0.8535	1.3996	0.9715	7.6820	0.8368	0.7404
DSIFT	74.5644	0.5289	0.8524	1.3666	0.9712	7.6797	<u>0.8415</u>	<u>0.7524</u>
CNN	74.6775	0.5406	0.8542	1.3745	0.9718	7.6796	0.8402	0.7516
U2Fusion	71.4127	0.6276	0.7940	1.6407	0.9735	7.5893	0.8221	0.6284
MUFusion	67.7763	0.6304	0.7847	1.6731	0.9675	7.6358	0.8219	0.6325
MFIF-GAN	74.5606	0.5294	0.8524	1.3666	0.9712	7.6796	0.8377	0.7462
ZMFF	74.3894	<u>0.6990</u>	0.8516	1.6281	0.9701	7.6847	0.8266	0.6950
IFCNN	<u>74.8327</u>	0.6845	0.8548	1.5829	0.9731	7.6851	0.8293	0.7252
SDNet	73.2805	0.6516	0.8308	1.6700	0.9790	7.6452	0.8240	0.6532
MFF-GAN	72.7623	0.6556	0.8253	<u>1.6774</u>	0.9723	7.6739	0.8237	0.6592
SwinFusion	72.3692	0.6913	<u>0.8576</u>	<u>1.6333</u>	0.9764	7.6974	0.8250	0.6931
FusionDiff	75.1462	0.7196	0.8633	1.6925	<u>0.9766</u>	<u>7.6853</u>	0.8446	0.7533

Table 4

The average scores of all algorithms on the MFFW dataset, where the highest and second highest scores are highlighted in bold and underlined, respectively.

Methods	<i>PSNR</i>	<i>Q_{AB/F}</i>	<i>SSIM</i>	<i>VIFF</i>	<i>CC</i>	<i>EN</i>	<i>Q_{NCIE}</i>	<i>Q_G</i>
NSCT-SR	72.1983	0.6312	0.7957	1.6255	0.9463	7.3634	0.8178	0.6239
GFF	72.2489	0.6307	0.8051	1.6547	0.9478	7.3348	0.8187	0.6246
CSR	71.9510	0.6219	0.7891	1.5971	0.9437	7.3457	0.8182	0.6151
DSIFT	71.9127	0.6304	0.7955	1.5872	0.9428	7.3404	0.8187	0.6245
CNN	72.0344	0.6281	0.8072	1.5930	0.9455	7.3261	<u>0.8196</u>	0.6222
U2Fusion	69.1245	0.5452	0.7608	1.6809	0.9588	7.1313	0.8169	0.5452
MUFusion	65.6914	0.5246	0.7411	1.7074	0.9525	7.0654	0.8167	0.5247
MFIF-GAN	71.8126	0.6265	0.7991	1.5784	0.9423	7.3237	<u>0.8196</u>	0.6207
ZMFF	71.7887	0.6618	0.8064	1.6741	0.9414	7.3133	0.8195	<u>0.6528</u>
IFCNN	<u>72.9156</u>	0.6071	<u>0.8173</u>	1.6890	0.9584	7.3268	0.8190	0.6026
SDNet	70.4461	0.5647	0.7943	1.6946	<u>0.9638</u>	7.2010	0.8181	0.5628
MFF-GAN	70.4173	0.5708	0.7892	<u>1.7075</u>	<u>0.9557</u>	7.3035	0.8177	0.5672
SwinFusion	71.1711	0.5996	0.7940	1.6893	0.9623	7.1806	0.8187	0.6003
FusionDiff	72.9973	<u>0.6516</u>	0.8422	1.7087	0.9648	<u>7.3478</u>	0.8214	0.6545

so the region should be consistent with (a). Contrary to expectations, the fusion results of U2Fusion(h), MUFusion(i), ZMFF(k), IFCNN(l), SDNet(m), MFF-GAN(n) and SwinFusion(o) show distortions in the clothing region, specifically, the presence of artifacts in these images. Similar to the above, as shown in the red boxes, the fusion results of CSR(e), DSIFT(f), and MFIF-GAN(n) show a block effect at the stick and the contours of the elderly face. In the above methods, the decision map-based methods adopt an averaging strategy at the junction of the clear and fuzzy regions, which causes a loss of detail information at the boundaries. And it is difficult for end-to-end methods to extract only the image features of the focused region, so that the fusion images are disturbed by the defocused features, which is a problem faced by all end-to-end MFIF methods. In addition, MUFusion(i) and SwinFusion(o) produce fused images that are significantly brighter than the source images, since it first converts RGB images into the YCbCr color space and performs image fusion only in the Y (luminance), which is actually a single-channel image fusion method. CNN(g) and FusionDiff(p) achieved relatively good fusion results in these comparison regions.

In summary, FusionDiff is independent of common fusion rules and adversarial game strategies, and can directly fuse RGB images without color space conversion, thus achieving the best fusion results in preserving more details and maintaining the same color distribution as that of the source images.

4.2.3. Computational complexity comparison

As shown in Table 5, we compare the parameters (size) and FLOPs (floating-point operations per second) of the proposed and other SOTA methods. The first pair of test images in the Lytro dataset is used to calculate FLOPs. Despite the large number of model parameters of the proposed method, it can still be trained on a single GPU, and the proposed method is competitive with some of the SOTA methods in terms of FLOPs.

4.3. Deterministic verification of fusion results

The certainty of fusion results is a fundamental principle for image fusion tasks. In other words, the fusion results obtained by repeated experiments with an image fusion method should be the same; otherwise, the optimal result cannot be determined. FusionDiff is essentially a conditional image generation model that generates clear fused images from random noise. Although the fused images generated by FusionDiff are not exactly the same for different input random noise, the difference between these images is so small that it can be ignored. Thus, it can be assumed that FusionDiff generates the same fused images according to different noises. We first verify this through experiments and then explain the reason.

In this subsection, four independent experiments are performed on a pair of images from the Lytro dataset; that is, fused images are generated based on four different random noises. Fig. 9 shows these fusion results and the corresponding residual images, and the same binarization operation is performed on the difference maps to show the details better. It can be seen that the four residual images are almost identical and it is difficult to observe the difference between them with the human eyes, which subjectively indicates that the fusion result of FusionDiff is unique for given source images.

In order to visualize the advantages of using Eq. (7) instead of Eq. (6) as the generation process of FusionDiff, Table 6 shows the evaluation metrics scores of four sets of fusion images generated from the same source images based on Eqs. (6) and (7). The Error of a group of fused images about a certain evaluation metric m is defined as $Error_m = (max_m - min_m)/mean_m$, and smaller error means the more stable fusion results. Fuse has less error and higher scores on each evaluation metric than Fuse-noise, suggesting that the Gaussian variance term in Eq. (6) not only leads to an uncertain generation process but also seriously affects the quality of the generated results. These objective results

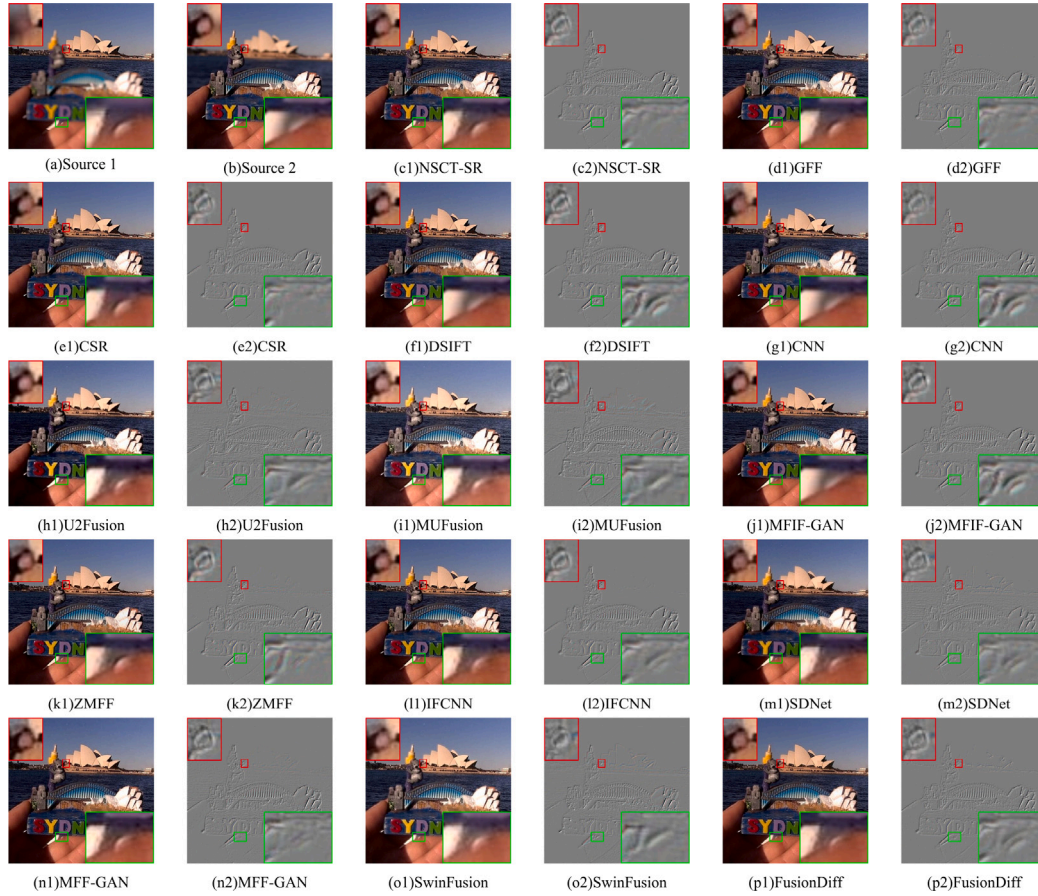


Fig. 7. The source and fused images of “Souvenir in front of Sydney Opera Hous” from the Lytro dataset obtained, where (c1)–(p1) denote the fused images and (c2)–(p2) denote the difference maps between the fused images and the source image (a), respectively.

Table 5

The comparison of total parameters and FLOPs for different methods.

Method	CNN	U2Fusion	MUFusion	ZMFF	IFCNN	SDNet	MFF-GAN	SwinFusion	MFIF-GAN	DDPM	FusionDiff
Size (M)	4.93	0.66	0.55	2.17	0.08	0.07	0.04	0.93	13.70	35.71	26.90
FLOPs (G)	106.82	356.64	49.65	77.68	35.25	36.36	21.70	271.10	1429.71	1514.80	239.86

Table 6

The evaluation metric scores of the four fusion results obtained using Eqs. (6) and (7), where Fuse denote the fusion results of FusionDiff based on Eq. (7) and Fuse-noise denote the fusion results of DDPM based on Eq. (6).

Metric	Fuse 1	Fuse 2	Fuse 3	Fuse 4	Error	Fuse-noise 1	Fuse-noise 2	Fuse-noise 3	Fuse-noise 4	Error
$PSNR$	76.91932	76.91921	76.91961	76.91924	0.0005%	67.15998	66.61408	67.06193	66.10432	1.5819%
$Q_{AB/F}$	0.72112	0.72116	0.72121	0.72117	0.0125%	0.62296	0.61673	0.62128	0.60871	2.3080%
$SSIM$	0.89862	0.89862	0.89863	0.89862	0.0011%	0.80676	0.79596	0.80469	0.78474	2.7593%
$VIFF$	1.68594	1.68594	1.68594	1.68597	0.0018%	1.66219	1.68173	1.66784	1.6747	1.1689%
CC	0.98789	0.98789	0.98789	0.98789	0	0.98520	0.98493	0.98516	0.98452	0.0696%
EN	7.32871	7.32872	7.32868	7.32867	0.0007%	7.27499	7.27139	7.27277	7.26771	0.1001%
Q_{NCIE}	0.83315	0.83314	0.83315	0.83314	0.0004%	0.82244	0.82283	0.82189	0.82298	0.1319%
Q_G	0.72774	0.72777	0.72774	0.72778	0.0061%	0.61736	0.62215	0.60926	0.62289	2.2063%

prove that FusionDiff has a highly deterministic and high-quality image fusion process. The reason for the certainty of FusionDiff is that it performs thousands of iterations, which makes the difference in the initial random noise have a very weak impact on the final fused image, which is determined by the guiding source images.

4.4. Noise robustness verification of the fusion model

Currently, the fusion performance of a vast majority of MFIF algorithms is verified through multi-focus datasets with complementary focused and defocused regions. Take the Lytro dataset (Nejati et al., 2015), where the focus regions of any pair of source images can be

stitched together to form an all-in-focus image, which is an idealized situation. In practice, the source images often contain unpredictable noise due to the impact of various factors on the imaging sensor and the interference of the transmission channel during the image transmission. However, most of the MFIF algorithms cannot address this non-ideal situation, and few studies have focused on this practical problem. FusionDiff can generate a clear fused image even in non-idealized cases where the source images contain unknown noise. This subsection will prove that FusionDiff is a noise-robust method by related experiments.

Among the SOTA methods in Table 2, transform domain methods and decision map-based deep learning methods retain all the original features of the source images during the fusion process. It is reasonable

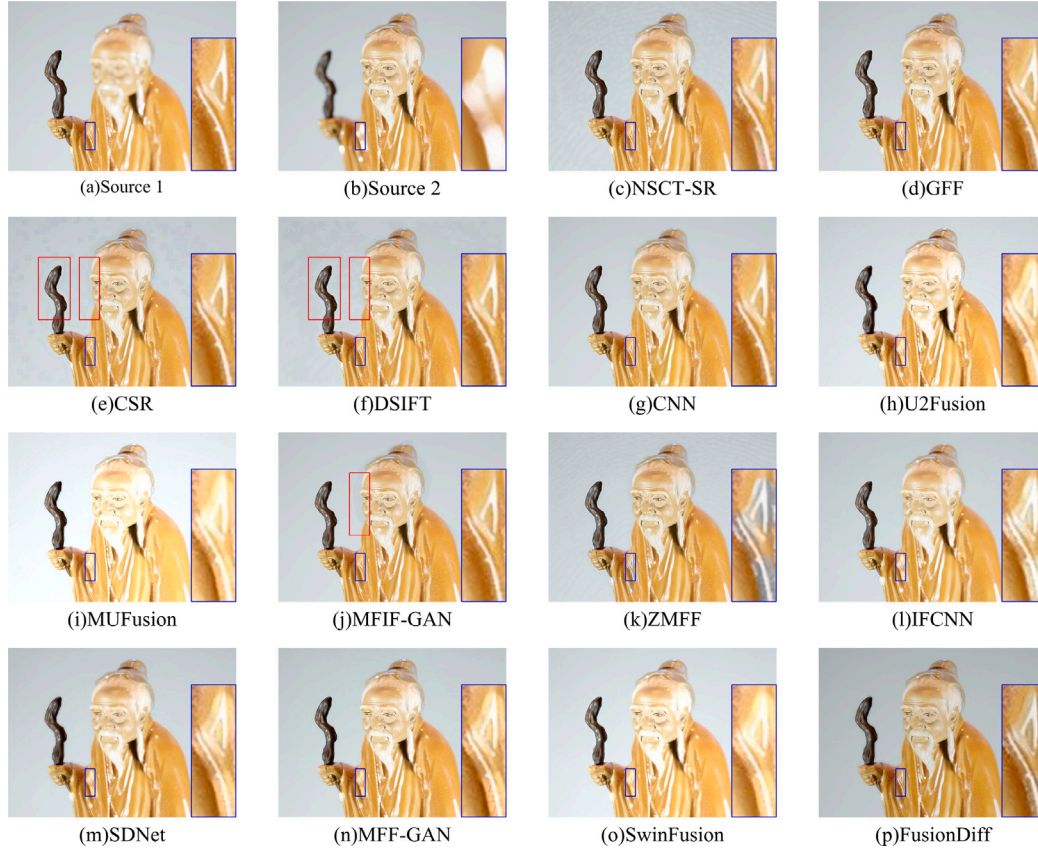


Fig. 8. The source and fused images of “old man with a stick” from the MFFW dataset.

to assume that these methods are not noise-robust, i.e., they will retain the noise in the source images intact in their fusion results. Only end-to-end deep learning methods have the potential to discard useless noise during the fusion process, as such methods may learn the ability to extract only valid information and ignore noise during training. To validate the idea, this paper compares the denoising capabilities of FusionDiff, IFCNN and SDNet, and the fusion results are assessed from both qualitative and quantitative evaluation perspectives.

4.4.1. Noise types and parameter settings

To verify the influence of noise on the fusion results, five common noise types, including Gaussian noise, impulse noise, Poisson noise, Rayleigh noise and uniform noise, are added to all source images in the dataset introduced in Section 4.1.1. The introduction and parameter settings of the five noise types are as follows. Uniformly, let I_s and I_n represent the noisy-free source image and the corresponding noisy source image, respectively.

- Gaussian noise is generated by blurring I_s based on a Gaussian filter (Zhang et al., 2020). The process of generating a blurred image I_n with Gaussian noise is expressed as:

$$I_n = G * I_s \quad (10)$$

where G is a Gaussian kernel of size 5×5 and $*$ denotes the convolution operation. G is defined as:

$$G(x, y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2+y^2}{2\sigma^2}} \quad (11)$$

where σ represents the standard deviation of G and $\sigma = 3$ in this paper.

- Impulse noise (Kayhan, 2014) is a random pixel point with two extreme values, 0 and 255, generated with the same probability. The process of generating I_n with impulse noise is expressed as:

$$I_n(x, y) = \begin{cases} 0, & \text{with probability } p \\ 255, & \text{with probability } (1 - p) \\ I_s(x, y), & \text{otherwise} \end{cases} \quad (12)$$

where p denotes the noise ratio and $p = 0.001$ in this paper.

- Poisson noise obeys the Poisson distribution (Deledalle, Tupin, & Denis, 2010), and its probability density function (PDF) is:

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad (13)$$

where λ denotes the expectation and variance of random numbers obeying the Poisson distribution, and $\lambda = 2$ in this paper.

- Rayleigh noise obeys the Rayleigh distribution (Kamboj et al., 2013), and its PDF is:

$$f(x) = \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}} \quad (14)$$

where σ represents the standard deviation of a set of random numbers randomly sampled from the Rayleigh distribution, and $\sigma = 5$ in this paper.

- Uniform noise obeys a uniform distribution (Boyat & Joshi, 2015), and its PDF is:

$$f(x) = \begin{cases} 1/(n - m), & m \leq x \leq n \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

where $\mu = (m + n)/2$ and $\sigma^2 = (n - m)^2/12$ denote the mean and variance, and $\mu = 1$ and $\sigma^2 = 5$, respectively.

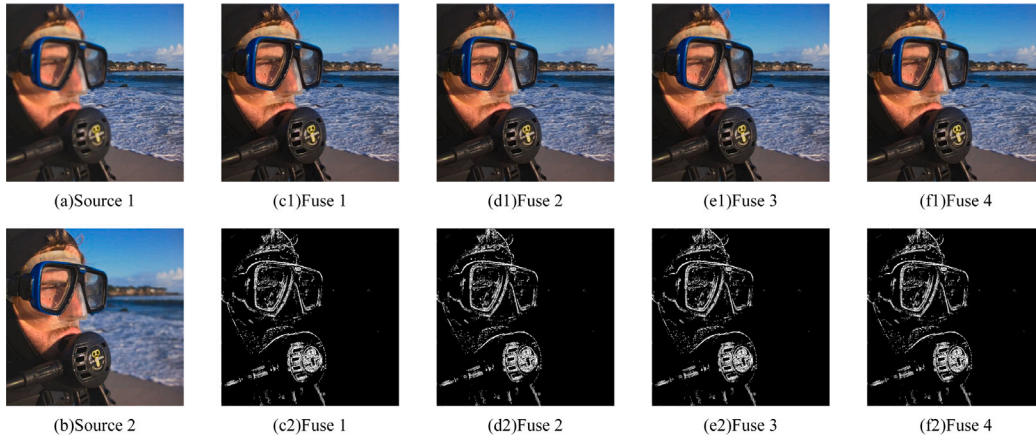


Fig. 9. Four fused images of a pair of source images from the Lytro dataset, where (c1)–(f1) are the fused images and (c2)–(f2) are the binarized difference maps between the fused images and the source image (a), respectively.



Fig. 10. The source images with five kinds of noise and the corresponding fused image, where (a1) and (b1) represent the noise-free source images, and (c1)–(e1) represent the corresponding fusion results. S1-Gauss represents S1 with Gaussian noise, IFCNN-Gauss represents the fusion result of S1-Gauss and S2-Gauss obtained by IFCNN, and so on.

4.4.2. Fusion results

As an extended experiment, instead of retraining the network, we use noisy source images to fine-tune the pre-trained model. Specifically, 20,000 epochs are used for training, and the other parameters are consistent with that in Section 4.1.2. Furthermore, IFCNN and SDNet are trained using source images with the same noise parameters to facilitate a fair comparison.

Fig. 10 shows the source images with five kinds of noise and the corresponding fused images for the last test image pair in the Lytro

dataset. The differences between the three sets of fusion results can be compared from the enlarged areas. All three methods showed a certain degree of denoising ability, but FusionDiff achieved the best fusion results for source images with different types of noise. There are blurring and noise in the fusion results of IFCNN and SDNet, especially (c3) is disturbed by the apparent impulse noise. It is worth noting that among all noises, these methods are most significantly affected by Gaussian noise. Since the blurred regions in the training dataset are obtained by applying Gaussian filtering to the clear images, it may

Table 7

The average fusion scores of three algorithms on the Lytro dataset. FusionDiff achieved the highest scores in all metrics except those underlined. The highest scores obtained by IFCNN and SDNet are bolded and correspond to the underlined numbers.

Noise type	<i>PSNR</i>	<i>Q_{AB/F}</i>	<i>SSIM</i>	<i>VIFF</i>	<i>CC</i>	<i>EN</i>	<i>Q_{NCIE}</i>	<i>Q_G</i>
IFCNN	74.8327	0.6845	0.8548	1.5829	0.9731	7.6851	0.8293	0.7252
IFCNN-Gauss	73.1565	0.4621	0.8351	1.5683	0.9740	7.4621	0.8213	0.6633
IFCNN-impulse	70.1940	0.6310	0.7235	1.6027	0.9325	7.5274	0.8234	0.6330
IFCNN-Possion	71.8589	0.6555	0.7858	1.6029	0.9429	7.5303	0.8166	0.6477
IFCNN-Rayleigh	73.7349	0.5852	0.8137	1.6054	0.9681	7.4729	0.8219	0.6152
IFCNN-uniform	73.0757	0.5986	0.7727	1.5953	0.9650	7.4769	0.8196	0.5988
SDNet	73.2805	0.6516	0.8308	1.6700	0.9790	7.6452	0.8240	0.6532
SDNet-Gauss	73.1138	0.4521	0.8277	1.6552	0.9773	7.4482	0.8137	0.5537
SDNet-impulse	73.2922	0.6607	0.7606	1.6611	0.9671	7.5503	0.8165	0.5927
SDNet-Possion	73.1340	0.6315	0.8216	1.6549	0.9766	7.5164	0.8184	0.6136
SDNet-Rayleigh	73.2350	0.5760	0.8254	1.6453	0.9760	7.4445	0.8229	0.6153
SDNet-uniform	73.1219	0.6042	0.8108	1.6599	0.9753	7.4863	0.8208	0.6146
FusionDiff	75.1462	0.7196	0.8633	1.6925	0.9766	7.6853	0.8446	0.7533
FusionDiff-Gauss	73.2761	0.5251	0.8433	1.6674	<u>0.9713</u>	<u>7.3824</u>	0.8229	0.5848
FusionDiff-impulse	74.4147	0.6692	0.8579	1.6920	<u>0.9715</u>	<u>7.4565</u>	0.8264	0.6706
FusionDiff-Possion	73.7790	<u>0.6514</u>	0.8566	1.6634	<u>0.9720</u>	7.5317	0.8554	0.6523
FusionDiff-Rayleigh	<u>73.5192</u>	0.6076	0.8505	1.6703	<u>0.9762</u>	7.4908	0.8239	<u>0.6084</u>
FusionDiff-uniform	73.3378	<u>0.5879</u>	0.8466	1.6922	0.9764	7.4865	0.8233	<u>0.5883</u>

lead to difficulties for the fusion model to learn the additional Gaussian noise information to remove it.

Table 7 lists the average scores of the corresponding fused images in the Lytro dataset for all types of source images. Particular, we use the noise-free source images and all fused images to compute the evaluation scores to fairly compare these fusion results and verify the noise robustness of the proposed method. Taking Fig. 10 as an example, for the fused images (c1)–(c6), (a1) and (b1) are used for calculation but (a2)–(a6) and (b2)–(b6) are not. Consistent with expectations, the highest scores of all eight metrics are obtained by the fused images of noise-free source images. Among the 40 metrics related to noise source images, FusionDiff obtained 31 highest scores, while SDNet and IFCNN received 9 and 2 highest scores, respectively. Although noisy source images have a certain impact on the fusion results, FusionDiff still achieves high scores on all evaluation metrics.

The demonstrated fusion results and comparison of objective evaluation metrics prove that FusionDiff can fuse source images with various noises into a noise-free fused image; that is, FusionDiff has certain noise robustness. Thinking in principle, since the image generation process of DDPM is a denoising process, FusionDiff can naturally remove the noise from the source images. In contrast, end-to-end methods like IFCNN and SDNet do not design special denoising modules and can only exhibit noise robustness with mediocre performance.

5. Conclusion and future work

A novel MFIF method based on DDPM is proposed in this paper. The fusion principle is to generate globally focused images from random Gaussian noise by iteratively performing multiple denoising operations under the guidance of two source images. The proposed approach has a simple network structure and fusion principle. It overcomes many difficulties existing MFIF methods face, including designing activity level measurements, fusion rules, feature extraction networks and training fusion networks. The contributions of this work include the following aspects: (1) The DDPM is introduced into the image fusion task for the first time, and an MFIF method, FusionDiff, based on DDPM is proposed. The experimental results show that FusionDiff obtains good quantitative and qualitative evaluation results. (2) Since FusionDiff is independent of any specific activity level measurement methods and fusion rules or complex feature extraction networks, it is more effective in complex MFIF tasks, such as those for multi-focus image pairs where focus and defocus regions appear to overlap. (3) FusionDiff is an image fusion method that supports few-shot learning. This advantage may make FusionDiff applicable to image fusion fields with scarce samples, such as microscopic images. (4) It is proven that FusionDiff has certain

noise robustness, and its fusion result is not disturbed by noise in the source images.

This paper demonstrates the feasibility of DDPM in the MFIF tasks. However, the mathematical nature of DDPM causes it to require thousands of iterations to obtain the final result, which results in a slow image fusion speed for FusionDiff. So we plan to use a more efficient noise variance function in the forward process and reduce the total diffusion step to perform the image fusion process faster. Meanwhile, in subsequent research, we will extend the versatility of the proposed method and validate the applicability of FusionDiff in other tasks, for instance, medical and multi-exposure image fusion.

CRedit authorship contribution statement

Mining Li: Conceptualization, Methodology, Software, Experiment, Writing – original draft. **Ronghao Pei:** Methodology, Experiment, Writing – review & editing, Supervision. **Tianyou Zheng:** Methodology, Writing – review & editing. **Yang Zhang:** Writing – review & editing, Funding acquisition. **Weiwei Fu:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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